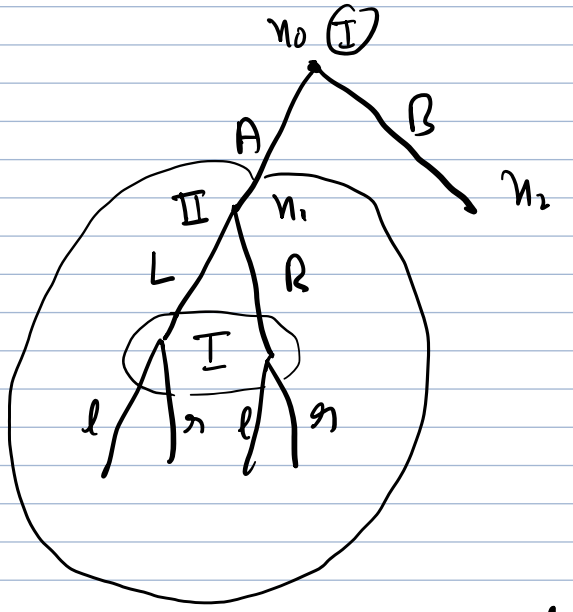
Class 28

Γ $\quad$ s $\quad$ Γ_n

$s: H_i \to \bigcup_{h \in H_i} A(h)$ $\quad$ s.t. $s(h) \in A(h)$ $\forall h \in H_i$

$b_i : H_i \to \bigcup_{h \in H_i} \Delta A(h)$ $\quad$ s.t $s(h) \in \Delta A(h)$ $\forall h \in H_i$

$\sigma_i : S_i \to [0,1]$ $\quad$ s.t. $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$



$\sigma_i(Al) = 0.5$
$\sigma_i(Ar) = 0.1$
$\sigma_i(Br) = 0.3$
$\sigma_i(Bl) = 0.1$

$\sigma_i'(l) = 0.5$
$\sigma_i'(r) = 0.5$

$\sigma_i(A)$, $\sigma_i(B)$ $\quad$ σ_i restricted Γ_{n_1}

$\sigma_i(l)$, $\sigma_i(r)$ $\quad$ $\sigma_i(l) = 0.6$

$\sigma_i'(l)$, $\sigma_i'(r)$ $\quad$ $\sigma_i(r) = 0.4$.

$\bar{\sigma}_i = \begin{cases} \sigma_i & \text{outside } \Gamma_n \\ \bar{\sigma}_i & \text{inside} \end{cases}$

$\bar{\sigma}_i(A) = \sigma_i(A)$ $\quad$ $\bar{\sigma}_i(l) = \sigma_i'(l)$

$\bar{\sigma}_i(B) = \sigma_i(B)$ $\quad$ $\bar{\sigma}_i(r) = \sigma_i'(r)$

Theorem Suppose Γ is an extensive form game and σ is an NE of the game. Further for a subgame Γ_n, suppose $p(n;\sigma) > 0$. Then σ induces an NE in the subgame Γ_n.

Proof

Suppose σ restricted to Γ_n is not an NE

$\exists$ i and a strategy σ_i' in the subgame Γ_n

s.t. σ_i' gives player i strictly more utility

You can construct a new strategy

$$\bar{\sigma}_i = \begin{cases} \sigma_i & \text{outside } \Gamma_n \\ \sigma_i' & \text{inside } \Gamma_n \end{cases}$$

As $\rho(n, \sigma) > 0$, we will have $\rho(n; (\bar{\sigma}_i, \sigma_{-i})) > 0$

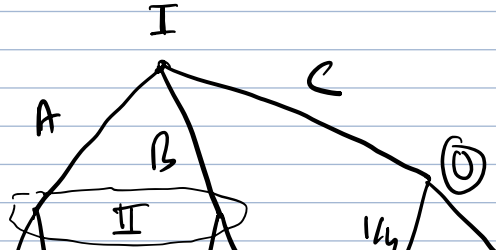
from there we can show that

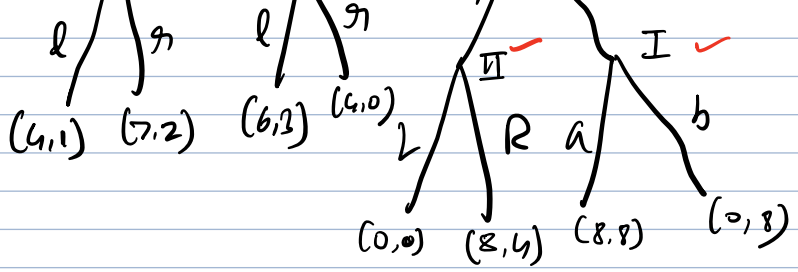
$$V_i(\sigma) < V_i(\bar{\sigma}_i, \sigma_{-i})$$

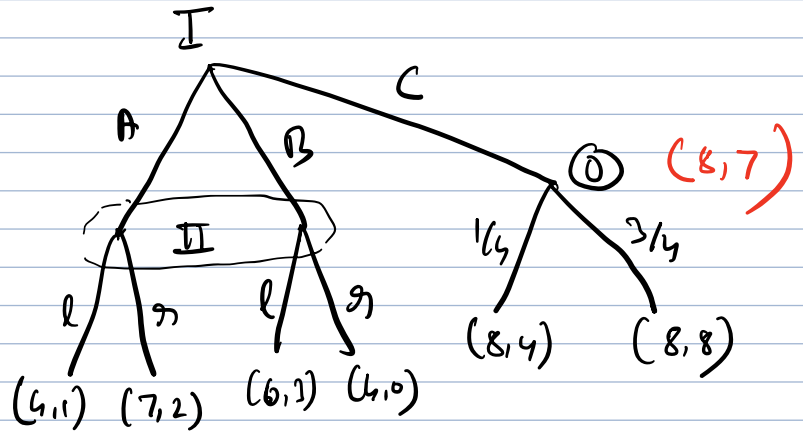
but contradicts

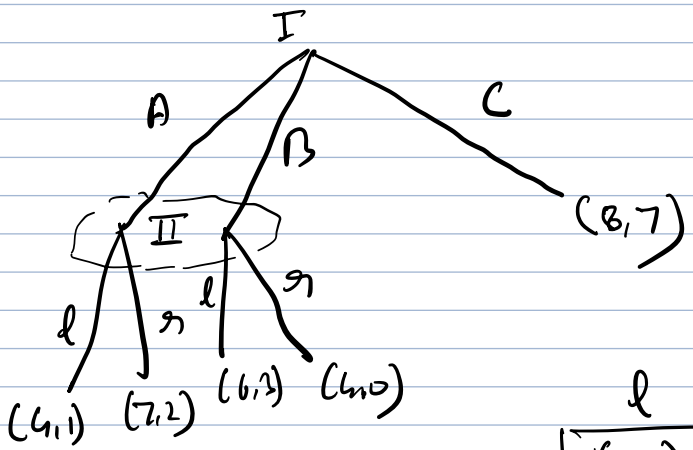
The full proof can be found in Solan's book.

Corollary Suppose σ is an NE of a game and under σ, all strategies of all players get +ve probabilities. Further suppose, all the chance nodes are giving +ve probabilities to all the edges coming out of the chance nodes. Then σ is an SPE.









find an NE of the game

	l	r
A	(4,1)	(7,2)
B	(6,3)	(4,0)
C	(8,7)	(8,7)

(C, l), (C, r) both are NE

(Ca, lR) and (Ca, rR) are the backward Induction equilibria of the game.

R-1 Every backward induction equilibrium is an SPE.

R-2 For any extensive form game there always exists an SPE.

Proof: We will discuss in the next class.